Sharp logarithmic complexity of adaptive continuous measurements

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Abstract

An antipodal-genus argument shows that n adaptive continuous real measurements have infinite worst-case error on $\mathbb{R}^m$ whenever $m \geq 2^n$. Combined with the known upper bound, this yields $\lfloor \log_2 m \rfloor + 1 \leq N(m) \leq \lceil \log_2 m \rceil + 1$ and hence $N(m) \sim \log_2 m$.

Scope and status. This establishes the sharp asymptotic growth expressly allowed as a target in AIM 553. It does not determine every exact value; in particular N(3) remains open. This is a solution claim submitted for review. Pre-submission mathematical, bibliographic, and editorial checks were performed with OpenAI Codex (AI); independent human review and formal verification are not claimed.

1 Problem and proof

For $m \geq 2$, let $N(m)$ be the least number of exact real measurements sufficient, for every $\varepsilon > 0$, to recover every $x \in \mathbb{R}^m$ with worst-case Euclidean error at most ε. We retain precisely the information model of entry 553 [1]: each branch functional is continuous in the input, while the dependence on the preceding exact measurements and the final decoder may be arbitrary. The question is discussed in Krieg–Ullrich [3], Section 10.3; the improved known upper bound is Theorem 1 of Krieg–Novak–Ullrich, version 2 [2].

Theorem 1.1. *For $m \geq 2$,*

$$\boxed{\lfloor \log_2 m \rfloor + 1 \leq N(m) \leq \lceil \log_2 m \rceil + 1.}$$

Consequently $N(m) = \log_2 m + O(1)$ and $N(m) \sim \log_2 m$. At powers of two, $N(2^r) = r + 1$ for $r \geq 1$.

More strongly, if $m \geq 2^n$, every deterministic algorithm using at most n adaptive continuous real-valued measurements has infinite worst-case error on $\mathbb{R}^m$.

1.1 An antipodal index and its elementary properties

For a nonempty compact antipodally invariant subset K of a Euclidean sphere, define its genus $\mathfrak{g}(K)$ to be the least integer $r \geq 1$ for which there exists a continuous odd map

$$K \longrightarrow \mathbb{R}^r \setminus \{0\}.$$

Set $\mathfrak{g}(\varnothing) = 0$. The Borsuk–Ulam theorem [4] implies

$$\mathfrak{g}(S_R^{m-1}) = m. \tag{1}$$

The upper bound uses the inclusion into $\mathbb{R}^m \setminus \{0\}$. A smaller odd map would contradict Borsuk–Ulam.

We will use three facts, proving the two fiber facts explicitly. First, an odd map from a compact symmetric subset $A \subset K$ to a sphere extends to an odd nonvanishing map on some invariant open neighbourhood of A in K. To see this, extend its coordinates to K by the Tietze extension theorem [5] (the compact metric space K is normal, and A is closed) and antisymmetrize:

$$F(x) = \frac{F_0(x) - F_0(-x)}{2}.$$

This agrees with the original map on A. By compactness it is nonzero on an invariant neighbourhood, where it can be normalized.

Lemma 1.2 (Odd zero sets)**.** *If $h : K \to \mathbb{R}$ is continuous and odd and $Z = h^{-1}(0)$, then*

$$\mathfrak{g}(Z) \geq \mathfrak{g}(K) - 1.$$

Proof. If $Z = \varnothing$, h itself gives $\mathfrak{g}(K) \leq 1$. Otherwise let $r = \mathfrak{g}(Z)$ and extend a nonvanishing odd map on Z to a continuous odd map $F : K \to \mathbb{R}^r$ by the preceding extension argument. The map

$$x \longmapsto (F(x), h(x))$$

is odd and nowhere zero: on Z its first component is nonzero, and outside Z its last component is nonzero. Thus $\mathfrak{g}(K) \leq r + 1$. □

Lemma 1.3 (Fibers of an even real function)**.** *If $f : K \to \mathbb{R}$ is continuous and even, some fiber satisfies*

$$\mathfrak{g}(f^{-1}(y)) \geq \left\lceil \frac{\mathfrak{g}(K)}{2} \right\rceil.$$

Proof. The empty-set case is immediate. For nonempty K, it suffices to show that if every fiber has genus at most r, then $\mathfrak{g}(K) \leq 2r$. Here necessarily $r \geq 1$. For each $y \in f(K)$, the compact invariant fiber $f^{-1}(y)$ has an invariant open neighbourhood U_y admitting an odd map to S^{r-1}. Compactness of K yields an open interval I_y around y with

$$f^{-1}(I_y) \subset U_y.$$

Indeed, otherwise a sequence outside U_y with $f(x_j) \to y$ would have a limit in $f^{-1}(y) \setminus U_y$.

Choose a finite interval cover of the compact set $f(K)$ from these I_y, and a Lebesgue number $\eta > 0$ for this relative cover. Take grid centres spaced by $d > 0$, with $6d/5 < \eta$, and intervals of radius $3d/5$ about those centres. These intervals cover the real line; finitely many cover $f(K)$. Give consecutive grid intervals alternating colors zero and one. Each interval, intersected with $f(K)$, lies in an original interval because its diameter is less than η. Intervals of the same color are pairwise disjoint.

Let $V_0, V_1 \subset K$ be the preimages of the two unions. They are invariant open sets covering K. On each component interval of a fixed color, use the odd map inherited from the relevant U_y. Since those component preimages are disjoint open sets, the maps together give continuous odd maps

$$v_j : V_j \to S^{r-1}, \qquad j = 0, 1.$$

For explicit even weights, put $\beta_j(x) = \operatorname{dist}(x, K \setminus V_j)$, with $\beta_j \equiv 1$ if $V_j = K$, and set $\alpha_j = \beta_j/(\beta_0 + \beta_1)$. The denominator is positive because V_0, V_1 cover K; these weights are continuous and even and vanish outside their respective V_j. Then

$$x \longmapsto (\alpha_0(x)v_0(x), \alpha_1(x)v_1(x))$$

with each product extended by zero outside its V_j is continuous, odd, and nonvanishing. Continuity of the zero extension follows because v_j is bounded and α_j vanishes on the boundary. This is the required map into $\mathbb{R}^{2r} \setminus \{0\}$. □

Lemma 1.4 (One continuous measurement)**.** *For any continuous $\lambda : K \to \mathbb{R}$, there exist $y \in \mathbb{R}$ and a compact invariant set $K' \subset K$ such that*

$$\lambda(x) = y \quad (x \in K'), \qquad \mathfrak{g}(K') \geq \left\lfloor \frac{\mathfrak{g}(K)}{2} \right\rfloor.$$

Proof. The function $h(x) = \lambda(x) - \lambda(-x)$ is odd. On its zero set Z, the restriction of λ is even. Apply Lemmas 1.2 and 1.3, obtaining

$$\mathfrak{g}(K') \geq \left\lceil \frac{\mathfrak{g}(K) - 1}{2} \right\rceil = \left\lfloor \frac{\mathfrak{g}(K)}{2} \right\rfloor.$$

The chosen fiber inside Z is compact and invariant. □

1.2 An adversarial transcript

With no measurements the error on $\mathbb{R}^m$ is infinite. Now fix any algorithm with $n \geq 1$ measurements, and suppose $m \geq 2^n$. An algorithm that sometimes stops earlier can be padded with zero measurements, so it is enough to consider exactly n stages. Fix an arbitrary radius $R > 0$ and start with $K_0 = S_R^{m-1}$.

Apply Lemma 1.4 to the first measurement. This chooses a value y_1 and a compact invariant set K_1 whose points all produce that value. For the second step, the preceding history is now fixed, so the algorithm selects one particular continuous function. Apply the same lemma to that function restricted to K_1. Continue for the first $n - 1$ stages. We obtain a fixed transcript $y_1, \ldots, y_{n-1}$ and a compact invariant set K_{n-1} producing it, with

$$\mathfrak{g}(K_{n-1}) \geq \left\lfloor \frac{m}{2^{n-1}} \right\rfloor \geq 2.$$

For the final, now fixed, branch functional λ_n, the odd function

$$x \longmapsto \lambda_n(x) - \lambda_n(-x)$$

has a zero on K_{n-1}; otherwise it would give an odd map into $\mathbb{R} \setminus \{0\}$, contradicting genus at least two. Hence some antipodal pair $x, -x$ of norm R produces the same complete transcript.

The decoder returns the same vector z on that transcript. By the triangle inequality,

$$\max\{\|x - z\|_2, \|-x - z\|_2\} \geq R.$$

Since R was arbitrary, the worst-case error is infinite. In particular, $n = \lfloor \log_2 m \rfloor$ cannot suffice. Combining the resulting lower bound with the cited upper bound proves Theorem 1.1.

Remark 1.5. The proof fixes one branch at a time. It never assumes continuity in the history, continuity of the whole adaptive encoder, or continuity of the decoder. The one-integer gap for non-powers of two remains; in particular this does not decide $N(3)$.

References

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[4] J. Matoušek, *Using the Borsuk–Ulam Theorem: Lectures on Topological Methods in Combinatorics and Geometry*, Universitext, Springer, 2003. doi:10.1007/978-3-540-76649-0.

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